

Problem Set 4

This abbreviated pset is due at 11am on **Wed Oct 5**. This is the **last pset covered on exam 1**.

Problem 1 (10+10 points)

Consider the equation $ABx = b$, given by:

$$\underbrace{\begin{pmatrix} 2 & 1 & 2 \\ & -1 & 2 \\ & & 1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} -1 & 1 & 2 & 1 \\ 3 & -2 & -5 & -1 \\ 0 & 1 & 1 & 2 \end{pmatrix}}_B x = \underbrace{\begin{pmatrix} 17 \\ 19 \\ 7 \end{pmatrix}}_b$$

- (a) Give **bases for the nullspaces** $N(A)$, $N(B)$, and $N(AB)$. (Hint: you should only have to do Gaussian elimination *once*.)
- (b) Give the **complete solution** x to $ABx = b$, and show how you can do this *without* every multiplying A and B together to explicitly form the matrix AB .

Problem 2 (10+5+5 points for A , b_1 , and b_2)

$Ax = b_1$ has **no** solution, and the **complete solution** to $Ax = b_2$ is:

$$x = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + \alpha_1 \begin{pmatrix} 1 \\ 1 \\ 0 \\ -1 \end{pmatrix} + \alpha_2 \begin{pmatrix} -2 \\ 0 \\ 1 \\ -2 \end{pmatrix}$$

for all possible scalars α_1 and α_2 .

Give **one** example of a possible matrix A (with **all nonzero entries**) and possible right-hand sides b_1 and b_2 (with any entries you want) that are consistent with the above facts. (The *same* A for both b_1 and b_2 .)

Hint: start by thinking about possible sizes and ranks for A , and then think how the columns are/aren't linearly dependent.

Problem 3 (5+5+10 points)

Consider the vectors:

$$a_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 1 \\ 1 \\ 5 \\ 4 \\ 9 \end{pmatrix}, \quad a_4 = \begin{pmatrix} 1 \\ 3 \\ 1 \\ 4 \\ 1 \end{pmatrix}.$$

Give **a basis** for the **orthogonal complement** of the subspace (of $\mathbb{R}^5$) **spanned by**:

(a) a_1

(b) a_2

(c) a_1, a_2, a_3 , and a_4 .

(Hint: this is a basis for $C(__)^\perp = __$.)